Weichen Wang — Curriculum Vitae

Contact Information

Department of Physics,
University of Milano-Bicocca
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Education

Johns Hopkins University, Baltimore MD, United States — 9/2016 - 12/2022
Ph. D. in Astrophysics, Department of Physics and Astronomy
Thesis Advisor: Susan Kassin (STScI); JHU Advisor: Timothy Heckman
Thesis title: *Observational studies of the relation between galactic winds and star formation at half the age of the Universe*

Tsinghua University, Beijing, China — 8/2012 - 7/2016
B. Sc. in Physics, Department of Physics
Thesis Advisor: Shude Mao

Research Positions

University of Milano-Bicocca, Milan MI, Italy — 12/2022 - now
Postdoc researcher, Department of Physics
Advisor: Sebastiano Cantalupo

University of California, Santa Cruz CA, United States — 2020; 7-9/2015
Visiting student, Department of Astronomy
Advisors: Sandra Faber, David Koo

Research Interests

Galaxy Formation in Extreme Large-scale Environments
Spectroscopic Studies of Galaxies and Their Circumgalactic Medium

Publications

Six leading-author papers published on Nature Astronomy, A&A, ApJ, and MNRAS (Five first-author papers and one student-led paper).
A total of 17 publications with 1000+ citations in past 3 years; h-index: 16.
See the attached publication list or this ADS link for full publication records.

Awards

Giovani Talenti (Young Talents Award)
by U. Milano-Bicocca & Accademia Nazionale dei Lincei of Italy, 2025

The International Astronomical Union Travel Grant, 2019

First-year student award, Johns Hopkins Physics & Astronomy Deptartment, 2016

National Astronomical Observatory of China Scholarship, 2016

Award of excellence by Tsinghua University: 清华学堂计划, 2013-2016

Media

Press releases by the **European Space Agency**, Nature Portfolio, U. Milano-Bicocca, Caltech, Swinburne U. (reporting W. Wang+ 2025): *"Big Wheel", a surprisingly large disk galaxy discovered in the early universe.*

News coverage by **Scientific American**, The Independent, Newsweek, Daily Mail, Il Giornale: *JWST Spots Giant Spiral Galaxy Shockingly Early in Cosmic History.*

News coverage by **Harvard Smithsonian** (in schedule for July 2026), Phys.org and others (reporting W. Wang+ 2026): *'Red Potato' galaxy discovered by astronomers.*

Service

NASA Astrophysics Program Review Panellist, 2026
SOC/LOC member, Conference What Matter(s) Around Galaxies, 2024
Referee for The Astrophysical Journal, Astronomy and Astrophysics

Talks

Invited Conference Talks:
The Controversial Fate of High-z Galaxy Protoclusters, Spain, 2025
Galaxies and Diffuse Gas in Large-Scale Overdense Environments, Italy, 2024

Contributed Conference Talks (selected from recent 3 years):
European Astronomical Society Annual Meeting, Switzerland, 2026
Zoom-in Views of Galaxy Disks Across Cosmic Time, Sweden, 2026
The Baryon Cycle from Reionization to the Cosmic Noon, Chile, 2025
Tracing Cosmic Evolution with Galaxy Clusters, Italy, 2025
Observing and Simulating Galaxy Evolution in the Era of JWST, Switzerland, 2024
What Matter(s) Around Galaxies 2024, Italy, 2024

Department Seminars:
Tsinghua University/Peking University/NAOC, China, 2023/2024
University of Missouri, U.S., 2020

Technical Expertise

JWST Spectroscopy and Imaging:
Serving as the deputy lead for JWST program GO1835, a spectroscopic survey of more than 50 galaxies in two $z \sim 3$ QSO fields; responsible for the planning, implementation, and analysis of the NIRSpec and NIRCam observations.

Keck & VLT spectroscopy: Reducing and analyzing the spectra of ≃200 z=1 galaxies as the member of a Keck/DEIMOS program HALO7D (P.I.: GuhaThakurta).

Observation Programs (Selected)

Summary: 100+ hours on JWST and 120+ hours on VLT & ALMA. Programs joined as P.I. and core member are listed and marked with [P] and [C], respectively.

[C] ALMA Cycle-12 Program (P.I.: A. Pensabene; 27 hours):
The Big Wheel Galaxy: a unique laboratory to study the physics of early disk formation in overdense environments, 2025-2026

[C] JWST Cycle-4 Program (P.I.: B. Wang; 20 hours):
Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features, 2025-2026

[C] VLT P112 Program (P.I.: S. Cantalupo; 84 hours): *Connecting the dots with MUSE: the Cosmic Web in emission around a massive structure at z=3,* 2024-2025

JWST Cycle-2 Program (P.I.: Kassin; 68 hr): *Galaxy angular momentum alignment with filaments at z∼3: The effect of large scale structure on galaxies,* 2024

[C] JWST Cycle-1 Program (P.I.: S. Kassin; 74 hours): *A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over 1<z<6,*, 2023

[C] JWST Cycle-1 Program (joined in 2022 with major contributions; P.I.: S. Cantalupo; 24 hours): *Unraveling the Knots of Gaseous Cosmic Web Filaments at z∼3 through H-alpha Emission Observations,* 2023

ALMA Cycle-8 Program (P.I.: R. Simons): *CO Kinematics at Cosmic Noon: Timing the Redistribution of Metals Around Galaxies,* 23.1 hours, 2022

[P] ALMA Cycle-7 Program (P.I.; 15 hours): *Does molecular gas follow the motion of ionized gas inside typical high-redshift star-forming galaxies?* Observations not completed due to weather and the impact of COVID-19 in Chile, 2021

Mentorship

Xiaohan Wang, Visiting Ph.D student of Tsinghua University, 2025
JWST NIRSpec studies of galaxy metallicity in the high-redshift cosmic web
Resulting in one research paper accepted to and in press at A&A.

M. Francesca Uboldi, Bicocca undergraduate in physics major, 2024 (thesis project)
Relations between galaxy colors and morphology with JWST medium-band imaging
Now pursuing Masters in Astronomy degree in Bicocca.

Ying Qin, Johns Hopkins undergraduate in physics, 2021-2023
Studying the Mg II emission and ionizing photons from low-mass galaxies at $z \sim 1$
Resulting in two conference poster presentations, including one at the AAS Annual Meeting; Now pursuing a career as lawyer in Washington D.C.

Teaching Experience

Volunteer Teaching Assistant, Laboratory of Data Analysis in Astrophysics
University of Milano-Bicocca, Spring 2023

Teaching Assistant, General Physics I for Biological Science Majors
Johns Hopkins University, Fall 2016

Teaching Assistant, General Physics Laboratory
Johns Hopkins University, Fall 2016

Outreach Activities

Science outreach day at University of Milano-Bicocca, 2024

A community event open to students from local primary/middle schools in Milan; co-designing the exhibition about astrophysical science with JWST

Member of the Astro Scholars program, Johns Hopkins University, 2021-2022

An annual program about astrophysics and coding for college students from underrepresented backgrounds; serving as a core member of the hiring & education team

Member of the Physics and Astronomy Graduate Students Outreach Team (PAGS), Johns Hopkins University, 2017-2019

Supporting visits of students from Baltimore local primary/middle schools around once per semester and teaching fundamental physics with educational demos

Volunteer teacher at the Pengzhai Primary School, Guizhou, China, Summer 2013

Teaching multiple STEM-related courses for Grade 3-6; the school, with very limited resources, is located in one of the least developed areas of the country.

Weichen Wang — Publication List

Summary

Six leading-author papers published in Nature Astronomy, A&A, ApJ, and MNRAS (Five as first author and one led by student).
A total of 17 publications with 1000+ citations in past 3 years; h-index: 16.
Full records are also available on the NASA ADS via this link.

Research topic categories marked below:
C1: Early-epoch galaxy formation in extreme cosmic environments; **C2:** Interplay between galaxies and the cosmic gas; **C3:** Galaxy star formation and dust.

First-Author Papers

[5,C1] **W. Wang**, S. Cantalupo, M. Galbiati, et al. arXiv: 2601.20473, A&A in press (2026): *A Quiescent Galaxy in a Gas-Rich Cosmic Web Node at z∼3*

[4,C1] **W. Wang**, S. Cantalupo, A. Pensabene, et al. Nature Astron., 9, 710 (2025): *A Giant Disk Galaxy Two Billion Years After The Big Bang* (Editorial Highlight)

[3,C2] **W. Wang,** S. A. Kassin, S. M. Faber, D. C. Koo, et al., ApJ, 930, 146 (2022): *The Baltimore Oriole's Nest: Cool Winds from the Inner and Outer Parts of a Star-Forming Galaxy at* $z = 1.3$

[2,C3] **W. Wang,** S. A. Kassin, C. Pacifici, et al., ApJ, 869, 161 (2018): *Galaxy Inclination and the IRX-β Relation: Effects on UV Star Formation Rate Measurements at Intermediate to High Redshifts*

[1,C3] **W. Wang,** S. M. Faber, F.-S. Liu, et al., MNRAS, 469, 4063 (2017): *UVI colour gradients of 0.4<z<1.4 star-forming main-sequence galaxies in CANDELS: dust extinction and star formation profiles*

Publications [4-5] have received extensive press coverages by major academic institutions/agencies and global news outlets.

Publications [1-3] are all cited and introduced by recent articles on the Annual Review of Astronomy and Astrophysics (ARA&A).

First-Author Papers in Prep.

[1,C2] **W. Wang,** S. A. Kassin, S. M. Faber, D. C. Koo, et al., in prep.: *The Roles of Integrated and Resolved Galaxy Star Formation in Driving Winds at* $z \sim 1$; Draft completed and to be submitted to ApJ in Spring 2026.

Papers with Leading Contributions

*[2,C1] X. Wang, S. Cantalupo, **W. Wang**, et al. A&A in press (2025): *Metal enrichment of galaxies in a massive node of the Cosmic Web at* $z \sim 3$

*paper led by PhD student co-mentored

[1,C1] A. Pensabene, S. Cantalupo, **W. Wang**, et al. A&A, 701, A120 (2025): *ALMA survey of a massive node of the Cosmic Web at z∼3: II. A dynamically cold disk galaxy in the proximity of a hyperluminous quasar*

Co-Author Publications (selected)

[16] Galbiati et al., A&A, 696, A95 (2025): *Connecting the growth of galaxies to the large-scale environment in a massive node of the Cosmic Web at z∼3*

[15] de la Vega et al., ApJ, 980, 168 (2025): *Improved SED-fitting Assumptions Result in Inside-out Quenching at z∼0.5 and Quenching at All Radii Simultaneously at z∼1*

[14] Travascio et al., A&A, 694, A165 (2025): *X-ray view of a massive node of the Cosmic Web at z∼3: I. An exceptional overdensity of rapidly accreting supermassive black holes*

[13] Liang, et al., ApJ, 986, 60 (2025): *Cosmic Himalayas: The Highest Quasar Density Peak Identified in a 10,000 deg2 Sky with Spatial Discrepancies between Galaxies, Quasars, and IGM H I*

[12] Ren et al., ApJ, 982, 200 (2025): *The Evolution of the Size and Merger Fraction of Submillimeter Galaxies across 1<z<6 as Observed by JWST*

[11] Pensabene et al., A&A, 684, A119 (2024): *ALMA survey of a massive node of the Cosmic Web at z∼3. I. Discovery of a large overdensity of CO emitters*

[10] Li et al., ApJS, 275, 27 (2024): *MAMMOTH-Subaru. II. Diverse Populations of Circumgalactic Lyα Nebulae at Cosmic Noon*

[9] Wang et al., MNRAS, 533, 2026 (2024): *SDSS-IV MaNGA: stellar rotational support in disc galaxies versus central surface density and stellar population age*

[8] de Beer et al., MNRAS, 526, 1850 (2023): *Resolving the physics of quasar Lyα nebulae (RePhyNe): I. Constraining quasar host halo masses through circumgalactic medium kinematics*

[7] Pérez-González et al., ApJL, 946, L16 (2023): *CEERS Key Paper. IV. A Triality in the Nature of HST-dark Galaxies*

[6] Pacifici et al., ApJ, 944, 141 (2023): *The Art of Measuring Physical Parameters in Galaxies: A Critical Assessment of Spectral Energy Distribution Fitting Techniques*

[5] Finkelstein et al., ApJL, 940, L55 (2022): *A Long Time Ago in a Galaxy Far, Far Away: A Candidate z∼12 Galaxy in Early JWST CEERS Imaging*

[4] Pharo et al., ApJS, 261, 12 (2022): *The Dwarf Galaxy Population at z∼0.7: A Catalog of Emission Lines and Redshifts from Deep Keck Observations*

[3] Simons et al., ApJ, 874, 59 (2019): *Distinguishing Mergers and Disks in High-redshift Observations of Galaxy Kinematics*

[2] Guo et al., ApJ, 853, 108 (2018): *Clumpy Galaxies in CANDELS. II. Physical Properties of UV-bright Clumps at 0.5<z<3*

[1] Liu et al., ApJL, 822, L25 (2016): *The UV-Optical Color Gradients in Star-forming Galaxies at 0.5<z<1.5: Origins and Link to Galaxy Assembly*